BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, LAHORE

INTERMEDIATE (12TH CLASS) EXAMINATION, 2025

CHEMISTRY

Past Paper 2025

Retyped faithfully from the original board paper — same questions, no changes.

GROUP – I

Objective Type

Time Allowed: 20 Min. | **Maximum Marks:** 17

Note: Four possible answers A, B, C and D to each question are given. The choice which you think is correct, fill that circle in front of that question with Marker or Pen ink in the answer book. Cutting or filling two or more circles will result in zero mark in that question.

Q1. Chlorine heptaoxide (Cl_2O_7) reacts with water to form:

(A) Hypochlorous acid
(B) Chloric acid
(C) Perchloric acid
(D) Chlorine and oxygen

Q2. Which of the following acid can be used as a catalyst in Friedel-Crafts reactions:

(A) $AlCl_3$
(B) HNO_3
(C) H_2SO_4
(D) HCl

Q3. The fibre which is made from acrylonitrile as monomer:

(A) PVC
(B) Rayon fibre
(C) Acrylic fibre
(D) Polyester fibre

Q4. Ozone layer is found in:

(A) Troposphere
(B) Stratosphere
(C) Hydrosphere
(D) Lithosphere

Q5. Mark the correct statement:

(A) Na^+ is smaller than Na atom
(B) Na^+ is larger than Na atom
(C) Cl^- is smaller than Cl atom
(D) Cl^- and Cl atom are equal in size

Q6. Cannizzaro's reaction is not given by:

(A) Formaldehyde
(B) Acetaldehyde
(C) Benzaldehyde
(D) Trimethyl acetaldehyde

Q7. Boric acid cannot be used:

(A) As antiseptic in medicine
(B) For washing eyes
(C) in soda bottles
(D) For enamels and glazes

Q8. Which compound will have maximum repulsion with H_2O:

(A) C_6H_6
(B) C_2H_5OH
(C) $CH_3CH_2CH_2OH$
(D) $CH_3–O–CH_3$

Q9. Ecosystem is a smaller unit of:

(A) Lithosphere
(B) Hydrosphere
(C) Atmosphere
(D) Biosphere

Q10. The state of hybridization of carbon atom in methane is:

(A) sp^3
(B) sp^2
(C) sp
(D) dsp^2

Q11. Which ion will have maximum value of heat of hydration:

(A) Na^+
(B) Cs^{+2}
(C) Ba^{+2}
(D) Mg^{+2}

Q12. Formula of chloroform is:

(A) CH_3Cl
(B) CCl_4
(C) CH_2Cl_2
(D) $CHCl_3$

Q13. The strength of binding energy of transition elements depends upon:

(A) Number of electron pairs
(B) Number of unpaired electrons
(C) Number of neutrons
(D) Number of protons

Q14. Elimination bimolecular reaction involves:

(A) First order kinetics
(B) Second order kinetics
(C) Third order kinetics
(D) Zero order kinetics

Q15. The brown gas formed, when metal reduce HNO_3 to:

(A) N_2O_5
(B) N_2O_3
(C) NO
(D) NO_2

Q16. A carboxylic acid contains:

(A) A hydroxyl group
(B) A carboxyl group
(C) A hydroxyl and carboxyl group
(D) A carboxyl and an aldehydic group

Q17. Which three elements are needed for the healthy growth of plants:

(A) N,S,P
(B) N,Ca,P
(C) N,P,K
(D) N,K,C

Essay Type

Time Allowed: 2.40 hrs. | **Maximum Marks:** 68

Note: SECTION – I

Q2. (Marks 16) Write short answers to any EIGHT (8) questions:

(i) Metallic character increases from top to bottom, justify.

(ii) The oxidation state of noble gas is usually zero, give reason.

(iii) Why 2% gypsum is added in cement?

(iv) Alkali metals are among the most reactive elements of the periodic table, give reason.

(v) Define vital force theory, how it was rejected?

(vi) What is natural gas? Write its two uses.

(vii) Alkanes are less reactive than alkenes, justify.

(viii) Write any two uses of methane.

(ix) Write structure of anthracene and biphenyl.

(x) Define aromatic hydrocarbon with one example.

(xi) Prepare propanoic acid using Grignard reagent.

(xii) S_N2 reactions are 2nd order reactions, justify.

Q3. (Marks 16) Write short answers to any EIGHT (8) questions:

(i) What is Chemical Garden?

(ii) Write down four uses of Al.

(iii) What is Aqua Regia, give one example?

(iv) Write down two dissimilarities of oxygen with sulphur.

(v) What is absolute alcohol, and how it can be obtained?

(vi) Write down reaction of phenol with formaldehyde in the presence of alkali.

(vii) Write down two methods for the preparation of acetaldehyde.

(viii) What is silver mirror test?

(ix) Write down the flavours of amyl butyrate and isobutyl formate esters.

(x) Write down the reaction of acetic acid with ammonia.

(xi) What is condensation polymerization, give one example?

(xii) What are acrylic resins, give one example?

Q4. (Marks 12) Write short answers to any SIX (6) questions:

(i) How municipal solid waste is incinerated? Discuss briefly.

(ii) What are chloramines? How are these formed?

(iii) Livestock waste is polluting water. Discuss.

(iv) Give the significance of ammonia fertilizer.

(v) Discuss the reactions taking place between 1 to 7 days during setting of cement.

(vi) Briefly discuss the colour chemistry in transition element compounds.

(vii) Give the formulae of: (a) Magnetite. (b) Haematite.

(viii) Give peculiar behaviour of fluorine.

(ix) Give formulae of: (a) Cryolite (b) Halite

> Note: SECTION – II

> Note: Note: Attempt any THREE questions.

Q5.

(a) [4] How do you justify the position of hydrogen at the top of group IA and VII A?

(b) [4] Describe with diagram the manufacture of sodium by Down's cell. Point out the three advantages of this process.

Q6.

(a) [4] Write down chain isomerism and functional group isomerism along with two examples each.

(b) [4] What are ortho para directing groups and meta-directing groups, give one example in each case?

Q7.

(a) [4] How will you bring about the following conversions: 2,2 (i) 1-Butene to 1-Butyne (ii) 1-Propanol to $CH_3–CH_2–CH_2–Cl$

(b) [4] Write a note on haloform reactions. Give the significance of iodoform test.

Q8.

(a) [4] Discuss Lucas test by giving suitable examples.

(b) [4] How will you prepare acetic acid by using: (i) Methyl magnesium bromide. (ii) Ethyl acetate.

Q9.

(a) [4] Give eight uses of Nobel gases.

(b) [4] Discuss rules for naming the complex compound of transition element by giving the examples.

GROUP – II

Objective Type

Time Allowed: 20 Min. | **Maximum Marks:** 17

Note: Four possible answers A, B, C and D to each question are given. The choice which you think is correct, fill that circle in front of that question with Marker or Pen ink in the answer book. Cutting or filling two or more circles will result in zero mark in that question.

Q1. The reaction between fat and NaOH is called:

(A) Esterification
(B) Hydrogenolysis
(C) Fermentation
(D) Saponification

Q2. β, β'-dichloroethyl Sulphide is commonly known as:

(A) Mustard gas
(B) Laughing gas
(C) Phosgene gas
(D) Bio-gas

Q3. Which is the strongest acid:

(A) $HClO$
(B) $HClO_2$
(C) $HClO_3$
(D) $HClO_4$

Q4. Ethanol can be converted into ethanoic acid by:

(A) Hydrogenation
(B) Hydration
(C) Oxidation
(D) Fermentation

Q5. Which of the following can be used as a catalyst in Friedel-Crafts reactions:

(A) $AlCl_3$
(B) HNO_3
(C) $BeCl_2$
(D) NaCl

Q6. Chile saltpetre has the chemical formula:

(A) $NaNO_3$
(B) KNO_2
(C) $Na_2B_4O_7$
(D) $Na_2CO_3.H_2O$

Q7. A single chloride free radical can destroy how many ozone molecules:

(A) 100
(B) 100,000
(C) 10,000
(D) 10

Q8. Co-ordination number of Pt in $[PtCl(NO_2)(NH_3)_4]$ is:

(A) 2
(B) 4
(C) 1
(D) 6

Q9. Which of the following is not a fatty acid:

(A) Propanoic acid
(B) Acetic acid
(C) Phthalic acid
(D) Butanoic acid

Q10. Mark the correct statement:

(A) Metallic character increases down the group
(B) Metallic character increases from left to right along a period
(C) Metallic character remains the same from left to right along a period
(D) Metallic character remains the same down the group

Q11. When CO_2 is made to react with ethyl magnesium iodide, followed by acid hydrolysis, the product formed is:

(A) Propane
(B) Propanoic acid
(C) Propanal
(D) Propanol

Q12. Tincal is a mineral of:

(A) Al
(B) B
(C) Si
(D) C

Q13. For which crop, ammonium nitrate fertilizer is not used:

(A) Cotton
(B) Wheat
(C) Sugarcane
(D) Paddy rice

Q14. Cannizzaro's reaction is not given by:

(A) Formaldehyde
(B) Acetaldehyde
(C) Benzaldehyde
(D) Trimethyl acetaldehyde

Q15. The pH range of the acid rain is:

(A) 7 – 6.5 (B) 6.5 – 6

(C) 6 – 5.6 (D) Less than 5

Q16. Select from the following the one which is alcohol:

(A) CH_3CH_2OH (B) $CH_3–O–CH_3$

(C) CH_3COOH (D) $CH_3–CH_2–Br$

Q17. Which of the following species has the maximum number of unpaired electrons:

(A) O_2 (B) O_2^+

(C) O_2^- (D) O_2^{2-}

Essay Type

Time Allowed: 2.40 hrs. | **Maximum Marks:** 68

Note: SECTION – I

Q2. (Marks 16) Write short answers to any EIGHT (8) questions:

(i) Define electron affinity. Give its trend within group and period of the periodic table.

(ii) What are halides? Give brief description of ionic halides.

(iii) Give two main points showing the difference of Beryllium from other 11A group elements.

(iv) Briefly explain the solubility of alkaline earth metal oxides in water.

(v) Briefly describe the functional group isomerism with at least two examples.

(vi) How is octane number of gasoline improved?

(vii) Write down the names of the following compounds according to IUPAC system: (a) $(CH_3CH_2)_3CH$ (b) $(CH_3)_2C=CH_2$

(viii) Give reactions showing the regeneration of acetylene from copper and silver acetylides.

(ix) What is the difference between ortho-para and meta directing groups?

(x) How is benzoic acid prepared from methyl and ethyl benzene?

(xi) How is n-Butane prepared through Wurtz Synthesis?

(xii) How is 1-Butanol prepared from ethyl magnesium bromide?

Q3. (Marks 16) Write short answers to any EIGHT (8) questions:

(i) Complete and balance the following chemical equations. (a) $HNO_3 + HI \rightarrow$ (b) $P + NO \rightarrow$

(ii) Give the reaction of H_2SO_4 (conc.) with: (a) C(s) (b) S(s)

(iii) Why is CO_2 gas at room temperature while SiO_2 is a solid?

(iv) Outline principal uses of borax.

(v) How will you differentiate chemically between methyl alcohol and ethyl alcohol? Give reactions.

(vi) Give the reaction of diethyl ether with HI.

(vii) What is the Tollen's test? Give its importance.

(viii) Give any four uses of acetaldehyde.

(ix) Write down formulas of palmitic acid and stearic acid.

(x) How does carboxylic acids react with carbonate and bicarbonate? Give reactions.

(xi) What is iodine number? Give its importance.

(xii) Briefly discuss the importance of enzymes (Any Two).

Q4. (Marks 12) Write short answers to any SIX (6) questions:

(i) Iodine is very important for pharmaceutical industry, give reason.

(ii) Write two methods for the preparation of xenon oxyfluorides.

(iii) Why d block elements show variable valency/oxidation states?

(iv) Write chemistry of chromyl chloride test.

(v) What is the meaning of setting of cement?

(vi) Write screening process in paper industry.

(vii) Atmosphere maintain heat balance of earth, justify.

(viii) Oil spillage affects the marine life, justify.

(ix) What is the role of ozone in atmosphere?

> Note: SECTION – II

> Note: Note: Attempt any THREE questions.

Q5.

(a) [4] How do you justify the position of hydrogen at the top of group I A and group IV A?

(b) [4] What is the role of lime in industries? Mention any eight (8) points.

Q6.

(a) [4] What are the types of structural isomerism? Explain any four types.

(b) [4] Explain with Mechanism Sulphonation of Benzene.

Q7.

(a) [4] Write the reactions of ethene with the following: (i) HBr (ii) Br_2 (iii) O_3 (iv) HOCl

(b) [4] Discuss the reaction of ethanal and propanone with: (i) Phenyl hydrazine. (ii) Hydroxylamine.

Q8.

(a) [4] How ethanol prepared from: (i) Molasses. (ii) Starch.

(b) [4] Discuss the mechanism for following conversions: (i) Acetic acid into ammonia. (ii) Acetic acid into ethanol.

Q9.

(a) [4] Explain the preparation of bleaching powder by Beckmann's method with diagram.

(b) [4] Define with one example the following term: (i) Ligands. (ii) Co-ordination sphere. (iii) Substitutational alloy. (iv) Central metal atom.